Team Project Final Report：Ames Housing Dataset

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1 Business Understanding

1.1 Problem

Banks need reliable ways to assess mortgage risk. Traditional appraisals may be biased or outdated, leading to potential overvaluation of collateral.

1.2 Motivation

A data-driven estimate of fair market value helps banks avoid lending against overpriced homes and better manage credit risk.

1.3 Data Mining Role

By predicting house prices from property and neighborhood features, we provide a benchmark. The gap between predicted vs. actual sale price can serve as a risk signal to adjust loan terms.

2 Data Understanding

2.1 Data Source

Ames Housing Dataset (2907 records, 70+ features). Includes lot size, living area, rooms, quality ratings, neighborhood, and sale price.

2.2 Characteristics

Mix of numeric, categorical, and ordinal variables. Target: SalePrice.

2.3 Potential Bias

Limited to Ames, Iowa (not fully generalizable). Time window is narrow (2006–2010).

2.4 Preparation

Requires data cleaning, missing value imputation, skew correction (e.g., log SalePrice), and encoding categorical variables.

3 Data Preparation

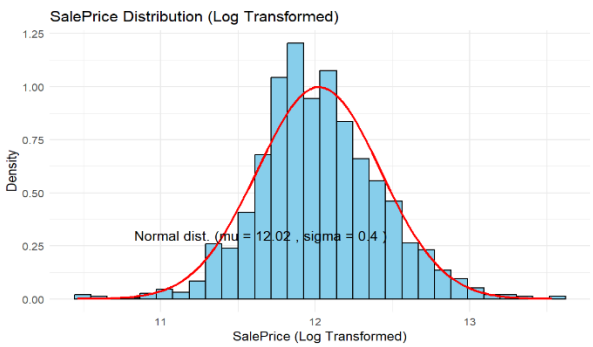
3.1 Data Integration

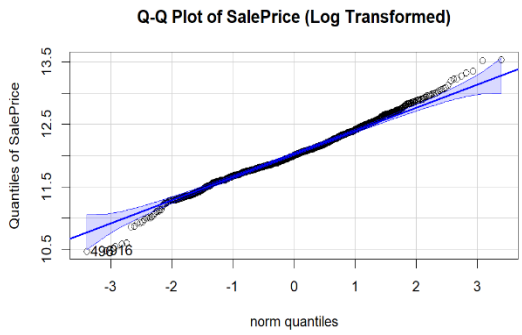
Combined the **training (1,460 rows)** and **test (1,447 rows)** sets to apply consistent preprocessing. Dropped the Id column since it provides no predictive value.

3.2 Outlier Handling:

Identified two extreme outliers in *GrLivArea* with unusually low prices and removed them to avoid distorting model training.

3.3 Target Transformation





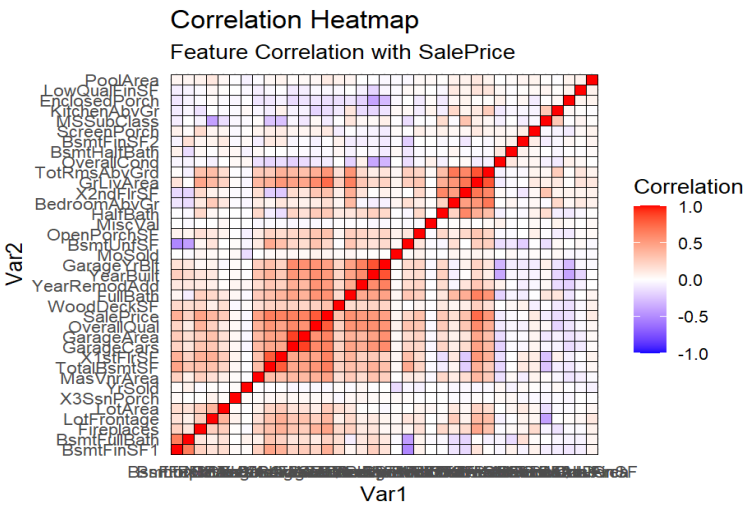
Applied log transformation to *SalePrice* to reduce right skewness and improve normality.

3.4 Missing Values

	Missing_Ratio
PoolQC	99.6902
MiscFeature	96.3855
Alley	93.2186
Fence	80.4819
FireplaceQu	48.6059
LotFrontage	16.6954
……	…..
Utilities	0.0344
Electrical	0.0344

Addressed systematically: features like *PoolQC, Alley, Fence, FireplaceQu* were filled with "None"; numerical structural features (e.g., *Basement, Garage* areas) were imputed with 0; *LotFrontage* was imputed by neighborhood median; categorical features were imputed with their mode. *Utilities* was dropped due to lack of variation.

3.5 Correlation Analysis



The heatmap revealed that variables such as *GarageCars* and *GarageArea*, as well as *TotalBsmtSF* and *GrLivArea*, were highly correlated, indicating potential redundancy.

Strong positive correlations with *SalePrice* were observed for *OverallQual*, *GrLivArea*, *GarageCars*, and *TotalBsmtSF*, highlighting the importance of property size and quality.

These findings support using both linear and nonlinear models — linear approaches to capture general trends, and tree-based models to handle complex feature interactions.

3.6 Feature Engineering

3.6.1 Converted certain numeric codes (*MSSubClass, YrSold, MoSold, OverallCond*) into **categorical variables**.

3.6.2 Applied **label encoding** to ordinal categorical variables (e.g., *ExterQual, KitchenQual, GarageFinish*).

3.6.3 Created a new feature ***TotalSF*** representing the sum of basement, first and second floor areas. (*Since area-related features play a significant role in determining house prices,*)

3.7 Skewness Reduction

Applied Box-Cox transformation to highly skewed numerical features.

3.8 Encoding

Generated dummy variables for all categorical predictors to ensure model compatibility.

3.9 Final Dataset

Produced a clean, numeric feature matrix with no missing values and consistent formatting. The final dataset includes transformed training and test sets ready for model fitting.

4 Modeling

We trained four regression models :

LASSO Regression ($\alpha = 1$, $\lambda = 0.0005$)

served as the **linear benchmark**, offering interpretability and reducing multicollinearity through regularization. It highlighted key predictors such as OverallQual, GrLivArea, TotalSF, and GarageCars.

Random Forest (500 trees)

captured nonlinear relationships and complex feature interactions, explaining **90.2% of the variance** and identifying OverallQual, TotalSF, and GarageArea as major drivers.

Gradient Boosting (n.trees = 3000, shrinkage = 0.05)

further improved predictive precision by iteratively **minimizing residual errors**, outperforming the Random Forest baseline.

XGBoost (nrounds = 1000, η = 0.05, max_depth = 4)

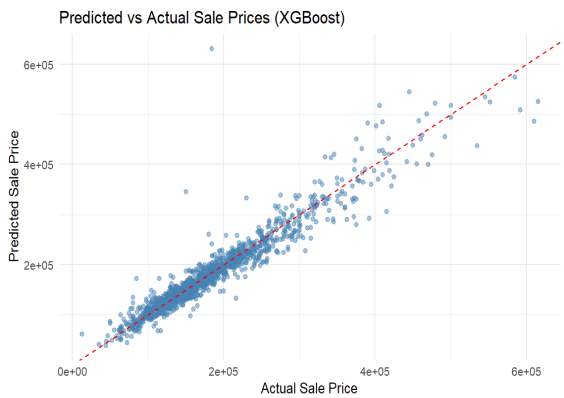
delivered the best trade-off between **accuracy and efficiency**, with a regularized boosting framework that enhanced stability and minimized overfitting.

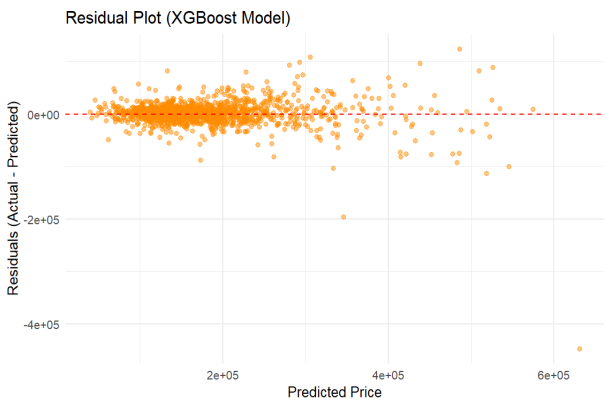
5 Evaluation

Model performance was assessed on the test dataset using **RMSLE** and **R^2**.

All models achieved strong predictive accuracy ($R^2 > 0.89$), with **XGBoost** performing the best overall.

Model	RMSLE	R^2	Interpretation
LASSO	0.153	0.892	Interpretable linear baseline; identifies main drivers
Random Forest	0.136	0.908	Captures nonlinear patterns; stable performance
Gradient Boosting	0.124	0.906	Improves precision via residual correction
XGBoost	**0.120**	**0.915**	Best generalization and lowest prediction error





The *Predicted vs. Actual Sale Price* plot shows that most points cluster tightly along the 45° line, indicating accurate predictions across the full price range. The *Residual Plot* confirms no major systematic bias, with errors symmetrically distributed around zero. Minor underestimation of luxury homes reflects limited representation of high-end properties in the dataset.

While **XGBoost** delivered the highest predictive performance, the **LASSO regression model** provided valuable interpretability. It highlighted key drivers such as *OverallQual*, *GrLivArea*, and *TotalSF*, confirming that the model's findings align with real-world appraisal logic. This interpretability is critical for financial institutions that require transparency and justification behind model-driven lending decisions.

Feature importance from XGBoost further reinforces these insights, emphasizing *TotalSF*, *OverallQual*, *GarageCars*, and *GrLivArea* as the most influential predictors of property value — again reflecting the dominant role of property size, construction quality, and functional space in determining fair market value.

From a business perspective, this model provides a **quantitative benchmark for mortgage collateral valuation**.

By integrating predictive modeling into loan underwriting, banks can:

- Detect overvalued properties early,
- Standardize risk-based pricing, and
- Reduce reliance on subjective human appraisals.

Even modest improvements (e.g., reducing valuation error by 5–10%) could yield measurable decreases in credit losses or over-lending exposure across large mortgage portfolios.

6 Deployment

The final **XGBoost model** will be deployed as a **decision-support tool** within the bank's loan underwriting system.

For each mortgage application, the model predicts a fair market value based on property and neighborhood characteristics.

If the predicted value significantly deviates from the declared sale price, the loan can be flagged for further review or adjustment in loan-to-value ratio.

Integration

The model can be implemented through an API or embedded dashboard linked to existing risk management systems. Periodic retraining (quarterly or semi-annually) using new housing data will ensure it reflects market conditions.

Risks and Ethical Considerations

Data Bias: The model was trained on Ames, Iowa data (2006–2010), which may not generalize nationally. Future iterations should expand regional and temporal coverage.

Fairness: Automated valuation tools must supplement, not replace, expert appraisals to avoid potential discrimination or over-reliance on model output.

Model Drift: Market dynamics (e.g., inflation, local zoning changes) may degrade performance, requiring continuous monitoring and recalibration.

Mitigation Strategies:

1. Regular retraining and validation with updated data.
2. Human review for flagged or extreme cases.
3. Fairness and bias audits across geographic and demographic segments